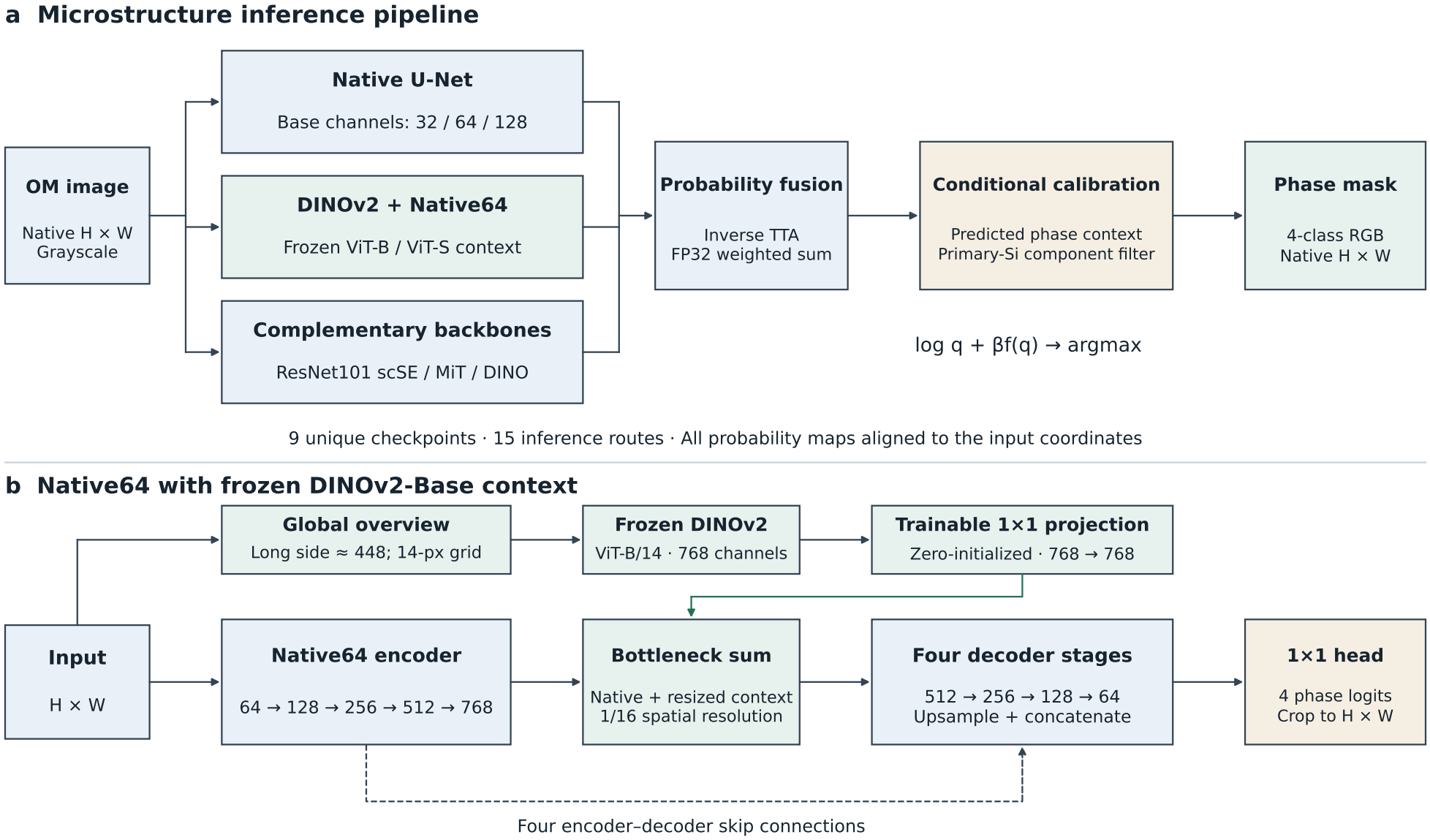
a Microstructure inference pipeline
OM image
Native H × W
Grayscale
Native U-Net
Base channels: 32 / 64 / 128
DINOv2 + Native64
Frozen ViT-B / ViT-S context
Complementary backbones
ResNet101 scSE / MiT / DINO
Probability fusion
Inverse TTA
FP32 weighted sum
Conditional calibration
Predicted phase context
Primary-Si component filter
log q + βf(q) → argmax
Phase mask
4-class RGB
Native H × W
9 unique checkpoints · 15 inference routes · All probability maps aligned to the input coordinates
b Native64 with frozen DINOv2-Base context
Input
H × W
Global overview
Long side ≈ 448; 14-px grid
Frozen DINOv2
ViT-B/14 · 768 channels
Trainable 1×1 projection
Zero-initialized · 768 → 768
Native64 encoder
64 → 128 → 256 → 512 → 768
Bottleneck sum
Native + resized context
1/16 spatial resolution
Four decoder stages
512 → 256 → 128 → 64
Upsample + concatenate
1×1 head
4 phase logits
Crop to H × W
Four encoder–decoder skip connections